

Anti-inflammatory and Antioxidant Effects of Japanese Herbal Medicine Kyoseihatekigan on Vocal Fold Wound Healing

*Satomi Ozawa, *Shigeyuki Mukudai, *Mami Kaneko, *Shota Kinoshita, *Keiko Hashimoto, *Yoichiro Sugiyama, †Subaru Hashimoto, †Junji Akaki, and *Shigeru Hirano, *Kyoto, and †Osaka, Japan

Summary: Objectives. The Japanese herbal medicine kyoseihatekigan (KHG) has been used to alleviate the symptoms of croaky voice and globus hystericus, and each of its components has anti-inflammatory and antioxidant effects. However, the mechanisms underlying these beneficial actions of KHG on the vocal folds remain largely unknown. We examined the effects of KHG on rat vocal fold wound healing and assessed its anti-inflammatory and antioxidant properties.

Study Design. Animal model.

Methods. The vocal folds of Sprague-Dawley rats were unilaterally injured under endoscopy. Rats were divided into three groups based on KHG dosing from pre injury day 4 to post injury day 3: 0 mg/kg/day (sham group), 500 mg/kg/day (1% KHG group) and 1000 mg/kg/day (2% KHG group). Histologic changes were examined to assess the degree of inflammation and oxidative stress at day 3, and fibrosis at day 56. In addition, gene expression related to pro-inflammatory cytokines and transforming growth factor-beta1 (TGF- β 1) signaling was examined by quantitative real-time polymerase chain reaction (qPCR).

Results. Histologic analysis showed that the 1% and 2% KHG treatments significantly decreased cell infiltration and the 4-hydroxy-2-nonenal-immunopositive area, and increased hyaluronic acid at day 3. Both KHG treatments significantly decreased fibrosis at day 56. qPCR revealed that mRNA of interleukin-1 β and cyclooxygenase-2 were significantly suppressed at day 1 and TGF- β 1 mRNA was significantly downregulated at day 5 in both KHG groups.

Conclusions. The current findings suggest that KHG has anti-inflammatory and antioxidant effects in the early phase of vocal fold wound healing, which can lead to better wound healing with less scar formation.

Key Words: Kyoseihatekigan—Vocal fold wound healing—Inflammation—Reactive oxygen species—Fibrosis.

INTRODUCTION

Vocal fold fibrosis causes permanent dysphonia that significantly reduces quality of life. The fibrosis is characterized by excessive accumulation of extracellular matrix, including type I collagen and decreased hyaluronic acid (HA), resulting in contraction and disruption of the layered structure of the lamina propria (LP).¹ There is still no definitive treatment for existing vocal fold fibrosis, and thus the prevention is critically important.

It is important to minimize the wound healing reaction,² including inflammation and oxidative stress, to prevent long-term scar formation. Fibrosis is preceded by an inflammatory response that begins immediately after injury or infection.³ This response is initially characterized by the release of pro-inflammatory mediators, typically tumor necrosis factor- α (TNF- α) and interleukin-1 β (IL-1 β), by

resident cells,³ and by the recruitment of inflammatory immune cells that subsequently release reactive oxygen species (ROS).⁴ Inflammation and ROS are necessary to successfully respond to the causative factor and promote tissue repair.^{3,4} However, exaggerated or prolonged inflammation and excessive ROS can lead to abnormal signaling that transforms the normal wound healing process into a profibrotic process.⁵⁻⁷ The above phenomena suggest that strict regulation of inflammation and oxidative stress is critical for prevention of vocal fold fibrosis. Indeed, Previous studies have been showed anti-inflammatory or antioxidant agents ameliorate vocal fold trauma or injury.^{8,9}

Kampo medicine, also known as Japanese traditional herbal medicine, has been used for centuries in Japan as a complementary, alternative treatment approach. It consists of extracts of multiple crude drugs, each with different targets, and it has the advantage of modulating multiple pathways simultaneously and producing synergistic effects in conditions with complex etiologies, like fibrosis.¹⁰ For instance, Sho-saiko-to (Xiao-Chai-Hu decoction) inhibits liver fibrosis via inhibition of the NF- κ B and transforming growth factor-beta1 (TGF- β 1) signaling pathways,¹¹ and due to its high quality and safety, the Japanese Ministry of Health, Labour and Welfare has approved it as a legal pharmaceutical product in Japan.¹² Furthermore, most modern Kampo medicines do not require preparation like traditional pharmaceuticals, and thus they are easily prescribed by physicians and have become popular and accepted in Japan.¹³

Accepted for publication October 11, 2021.

This study was supported by a Grant-in-Aid for Scientific Research-B (#19H03805) from The Ministry of Education, Culture, Sports, Science and Technology, a Grant-in-Aid for Early-Career Scientists (#19K18813) and a Grant from Kobayashi Pharmaceutical Co., Ltd. The authors have no other funding sources, financial relationships, or conflicts of interest to disclose.

From the *Department of Otolaryngology-Head and Neck Surgery, Kyoto Prefectural University of Medicine, Kyoto, Japan; and the †Kobayashi Pharmaceutical Co., Ltd., Osaka, Japan.

Address correspondence and reprint requests to Shigeyuki Mukudai, Department of Otolaryngology-Head and Neck Surgery, Kyoto Prefectural University of Medicine, Kajii-cho, Kamigyo-ku, Kyoto 602-8566, Japan. E-mail: s-muku@koto.kpu-m.ac.jp

Journal of Voice, Vol. 38, No. 2, pp. 503–509
0892-1997

© 2021 The Voice Foundation. Published by Elsevier Inc. All rights reserved.
<https://doi.org/10.1016/j.jvoice.2021.10.003>

TABLE 1.
Herbal Medicines Composing Kyoseihatekigan

	Crude Drug	
Forsythia fruit	2.5 g	(14.2%)
Platycodon root	2.5 g	(14.2%)
Glycyrrhiza	2.5 g	(14.2%)
Rhubarb	1.0 g	(5.7%)
Amomum seed	1.0 g	(5.7%)
Cnidium rhizome	1.0 g	(5.7%)
Myrobalan fruit	1.0 g	(5.7%)
Gambir	2.0 g	(11.4%)
Mentha herb	4.0 g	(22.8%)

Kyoseihatekigan (KHG) is a Kampo medicine that has been used to treat hoarse voice and throat discomfort due to laryngitis resulting from a cold or excessive singing or shouting.¹⁴ Japanese clinical study showed positive effects of KHG for patient with acute or chronic pharyngitis with no side effect.¹⁵ As defined by the Japan Kampo Medicines Manufacturer's Association, it is a mixture of extracts from the following nine ingredients: forsythia fruit, platycodon root, glycyrrhiza, rhubarb, amomum seed, cnidium rhizome, myrobalan fruit, mentha herb, and gambir (Table 1). Several investigators have suggested possible anti-inflammatory or antioxidant effects of these ingredients, but it is still unknown how KHG works on the vocal fold after injury. We examined the effects of KHG on vocal fold wound healing using a rat model by histological assessment and gene expression analysis related to inflammation and fibrosis.

MATERIALS AND METHODS

Experimental animals

Seventy-two male Sprague-Dawley rats aged 12-13 weeks were used in this study. They were purchased from Shimizu Laboratory Supplies Co. (Kyoto, Japan) and housed in plastic cages with standard bedding and free access to food and water. The temperature was maintained at 22°C with a 12-hour light/dark cycle. Rats were subjected to unilateral vocal fold injury and divided into three groups according to KHG dose: sham group, 500 mg/kg/day (1% KHG group), and 1000 mg/kg/day (2% KHG group). KHG was added to standard feed by Kobayashi Pharmaceutical Co. (Osaka, Japan). The KHG groups were fed the experimental diets *ad libitum* from 4 days before vocal fold injury to 3 days postoperatively, whereas the sham group received a normal diet throughout the study. The timing of the KHG diet was selected to cover the critical periods of early wound healing when inflammatory response and anti-oxidant effects peak.^{16,17} The rats were confirmed to normally get weight at each time point of sacrifice in all groups. All animal experiments were approved by the Animal Care and Use Committee of the Kyoto Prefectural University of Medicine and were conducted in accordance with its guidelines.

Surgical procedure

Vocal fold injury was surgically induced as established previously.¹⁸ Briefly, vocal folds were unilaterally injured using a 22-gauge needle to strip the vocal folds until the thyroarytenoid muscle was exposed under deep anesthesia with a mixture of three agents,¹⁹ specifically medetomidine hydrochloride (0.3 mg/kg), midazolam hydrochloride (4 mg/kg), and butorphanol tartrate (5 mg/kg), along with atropine (0.005 mg/kg) to reduce the secretion of saliva and sputum, and topical 4% lidocaine to reduce vocal fold movement. All rats did not receive any pain medicines after surgery. At each time point, animals were humanely sacrificed with intraperitoneal injection of pentobarbital sodium (200 mg/kg) and the larynges were harvested immediately.

Histological and immunohistochemical evaluation

The animals were humanely sacrificed at day 3 and 56 ($n = 6$ for each group at each time point). Whole larynges were dissected and immediately fixed in 4% paraformaldehyde for 4 hours, then immersed in 30% sucrose for 24 hours at 4°C and finally soaked in embedding medium (FSC22 Frozen Section Media; Leica Instruments, Nussloch, Germany) and frozen with liquid nitrogen. The block was cut into 8- μ m sections (CM 1850 Cryostat; Leica). Sections were mounted on glass slides, air dried, and stored at -80°C until staining.

The sections were stained with routine hematoxylin and eosin (H.E.; Merck KGaA, Darmstadt, Germany), Alcian blue pH 2.5 (AB; Abcam, Kenbridge, UK), and Elastica van Gieson (EVG; Merck KGaA). H.E. staining was used to evaluate the cell infiltration in the LP. The number of stained nuclei was counted at $\times 400$ magnification under Biorevo BZ-X700 microscope (Keyence, Osaka, Japan) at three different portions in the LP: the rostrocaudal mid portion, rostral, and caudal portions. BZ-X700 analyzer software automatically measured the area of specified colors in each section. AB staining and a hyaluronidase digestion technique were performed to identify HA, which is stained blue. EVG was used to identify collagen, which is stained pink. The percentage of the stained areas for each color in the LP was calculated as the density. Normalized ratios were obtained by dividing the percentage of each area on the injured side by that on the uninjured side.

4-hydroxy-2-nonenal (4-HNE), a marker of ROS, was examined at day 3 by immunohistochemistry. The sections were permeabilized in 0.1% Triton X-100 for 10 minutes then incubated with 2% albumin bovine serum (BSA, Nacalai Tesque, Kyoto, Japan), a blocking buffer, for 60 minutes, and incubated overnight at 4°C with rabbit anti-4-HNE antibody (Alpha Diagnostic International, San Antonio, TX; 1:500). After washing with phosphate-buffered saline, the slides were treated with goat anti-rabbit Alexa Fluor 488 secondary antibody (1:1000; Invitrogen, Carlsbad, CA) for 60 minutes. Nuclei were counterstained with 4'-diamidino-2-phenylindole. Experiments performed without the primary antibody comprised the negative

control. All images were acquired at $\times 200$ magnification, using identical settings for exposure and shutter speed, with a Biorevo BZ-X700 microscope (Keyence). BZ-X700 analyzer software (Keyence) was used to measure the 4-HNE-immunopositive areas using the same criteria as those involving measurements in the LP, and calculated the percentage of the immunopositive areas in the LP. Normalized ratios were calculated as a value relative to normal control vocal folds as previously described.⁹

The measurements were employed at the center of injured site at the midmembranous portion of the vocal folds using a BZ-X700 microscope (Keyence) by two observers who were unaware of the sample origins. Inter-rater reliability was confirmed by Pearson's correlation coefficient.

Quantitative real-time polymerase chain reaction (qPCR)

Gene expressions of inflammatory cytokines including IL-1 β , cyclooxygenase-2 (COX-2), and TNF- α were examined at day 1 after the injury by qPCR, because the cytokines increase immediately after the injury. Gene expressions of SMAD3, TGF- β 1, and type I procollagen were examined at day 5, because TGF- β 1 is reported to increase and peak around 7 days in rat vocal fold injury model. Six animals were used for each group at each time point.

After the whole larynges were harvested, they were immediately submerged in RNeasy Protect Tissue Reagent (Qiagen, Valencia, CA), then the epithelium and LP of the injured vocal folds were dissected away under the microscope. The RNeasy Mini Kit (Qiagen) was used to

purify total RNA according to the manufacturer's protocol. Total RNA quantification and quality were assessed by spectroscopy (NanoDrop ND-1000; Thermo Fisher Scientific, Waltham, MA). Then, total RNA was converted to complementary DNA (cDNA) using a High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Foster City, CA). The TaqMan Gene Expression Kit (Life Technologies, Grand Island, NY) and Lightcycler 480 Probe Master (Roche Diagnostics K.K., Basel, Switzerland) were employed for quantitative analyses. TaqMan primer probes for *Il1b* (IL-1 β , Rn00580432_m1), *Ptgs2* (COX-2, Rn01483828_m1), *Tnf* (TNF- α , Rn99999017_m1), *Smad3* (SMAD3, Rn00565331_m1), *Tgfb1* (TGF- β 1, Rn00572010_m1), *Col1a1* (type I procollagen, Rn01463848_m1), and succinate dehydrogenase complex, subunit A (*Sdha*) (SDHA, Rn00590475_m1) were employed. The $\Delta\Delta C_t$ method was employed with SDHA which previously validated²⁰ as the housekeeping gene for the determination of relative expression levels, and uninjured normal rats were used as a calibrator. No-template controls were performed with water instead of cDNA to ensure the absence of contamination.

Statistical analysis

Bartlett's test was used to assess the homogeneity of variances. One-way analysis of variance (ANOVA) with a post hoc Tukey test was performed for normally distributed data, whereas the Kruskal-Wallis test and the post hoc Steel-Dwass test were performed for non-normally distributed data. $P < 0.05$ was regarded as statistically significant.

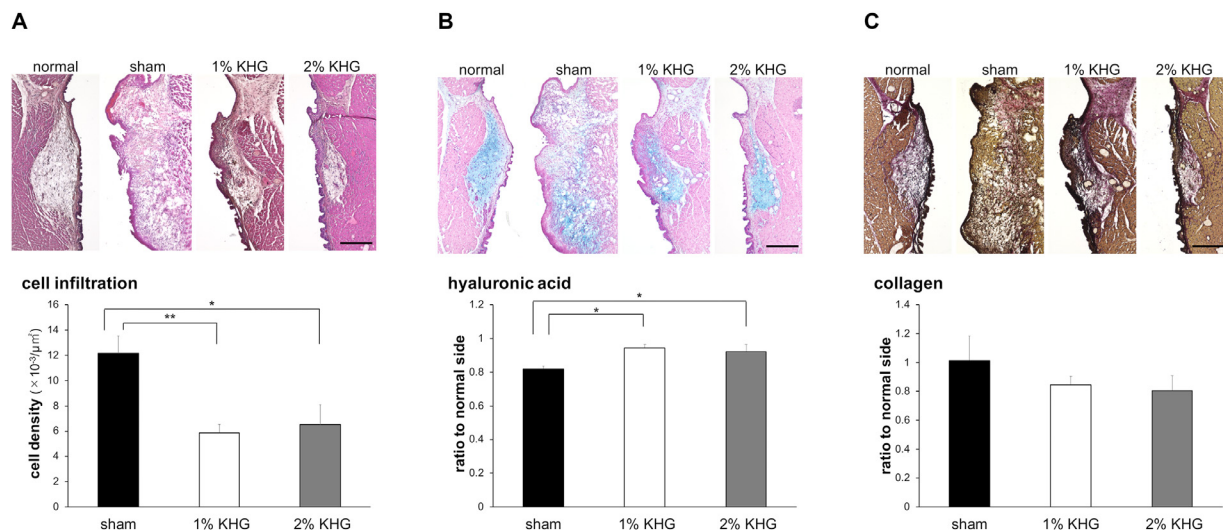


FIGURE 1. (A) Representative image of H.E. staining of rat vocal fold sections from a normal rat (uninjured side), sham group, 1% KHG group, and 2% KHG group on post injury day 3. The cell density is significantly suppressed in the 1% and 2% KHG groups compared to the sham group. (B) Representative image of AB staining in rat vocal fold from a normal rat (uninjured side), sham group, 1% KHG group, and 2% KHG group on post injury day 3. The normalized ratio of the HA-stained area is significantly increased in the 1% and 2% KHG groups compared to the sham group. (C) Representative images of EVG staining of rat vocal fold sections from a normal rat (uninjured side), sham group, 1% KHG group, and 2% KHG group on post injury day 3. There are no significant differences between groups in the normalized ratio of the collagen-stained area. $n = 6$ per group, $*P < 0.05$, $**P < 0.01$. Data are expressed as mean $\pm$ SEM. Scale bar = 200 μm . AB, Alcian blue; EVG, Elastica van Gieson; HA, hyaluronic acid; H.E., hematoxylin and eosin; KHG, kyoseihatekigan.

All data are expressed as the mean $\pm$ standard error of mean (SEM).

RESULTS

KHG reduced cell infiltration and restored HA on post injury day 3

The sham group showed massive cell infiltration and decreased HA on post injury day 3, whereas the 1% and 2% KHG groups demonstrated less cell infiltration and HA restoration (Figure 1A, B). Statistical analysis showed that the cell density was significantly reduced in the 1% and 2% KHG groups compared to the sham group ($P = 0.008$, $P = 0.016$, 1% and 2% KHG groups, respectively, Figure 1A), and the normalized ratio of the HA-stained area was significantly greater in the 1% and 2% KHG groups than in the sham group ($P = 0.015$, $P = 0.043$, 1% and 2% KHG groups, respectively, Figure 1B). There was no significant difference between groups in the normalized ratio of the collagen-stained area (Figure 1C).

KHG reduced 4-HNE expression on post injury day 3

In the sham group, the injured side exhibited greater 4-HNE expression than the uninjured side on post injury day 3 (Figure 2). The normalized ratio of the 4-HNE-immunopositive area was significantly lower in the 1% and 2% KHG groups than in the sham group ($P = 0.0007$, $P = 0.006$, 1% and 2% KHG groups, respectively, Figure 2).

KHG restored HA and reduced collagen deposition on post injury day 56

Fifty-six days after injury, the sham group exhibited a marked decrease in HA and dense collagen deposition (Figure 3A, B), both of which were significantly attenuated in the 1% and 2% KHG groups (HA, $P = 0.046$, $P = 0.046$, Figure 3A; collagen, $P = 0.028$, $P = 0.016$, 1% and 2% KHG groups, respectively, Figure 3B).

KHG altered gene expression of IL-1 β , COX-2, or TNF- α on post injury day 1

Both 1% and 2% KHG treatments significantly suppressed the mRNA of IL-1 β , COX-2 (IL-1 β , $P = 0.043$, $P = 0.043$, Figure 4A; COX-2, $P = 0.024$, $P = 0.024$, 1% and 2% KHG groups, respectively, Figure 4B). TNF- α mRNA was significantly lower in the 1% KHG group than in the sham group ($P = 0.038$, Figure 4C). It was also decreased in the 2% KHG group, but the difference did not reach significant as compared to sham.

KHG altered gene expression of TGF- β 1 on post injury day 5

Both 1% and 2% KHG treatments significantly suppressed TGF- β 1 mRNA ($P = 0.045$, $P = 0.045$, 1% and 2% KHG groups, respectively, Figure 4E). SMAD3 and type I

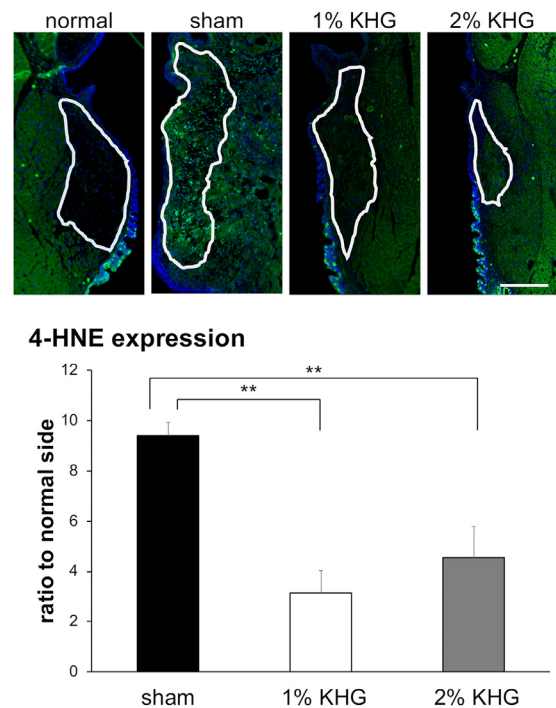


FIGURE 2. Representative images of immunostaining for 4-HNE (green) and DAPI (blue) in rat vocal fold sections from a normal rat (uninjured side), sham group, 1% KHG group, and 2% KHG group on post injury day 3. The normalized ratio of the immunopositive area for 4-HNE is significantly reduced in the 1% and 2% KHG groups compared to the sham group. $n = 6$ per group, $**P < 0.01$. Data are expressed as mean $\pm$ SEM. Scale bar = 200 μ m. In each image, the dotted white line encloses the lamina propria of the vocal fold. DAPI, 4'-6-diamidino-2-phenylindole; KHG, kyoseihatekigan; ROS, reactive oxidant species; 4-HNE, 4-hydroxy-2-nonenal. "Color version available online."

procollagen mRNA were also decreased, but the difference did not reach statistical significance (Figure 4D, F).

DISCUSSION

The prevention of vocal fold fibrosis remains a significant clinical challenge. To the best of our knowledge, this is the first study to demonstrate that KHG ameliorated vocal fold fibrosis by regulating both inflammation and ROS in a rat vocal fold injury model.

KHG consists of multiple ingredients (Table 1). Mentha herb, platycodon root, and glycyrrhiza have been widely used for throat disorders in herbal medicine,²¹ and a clinical study showed that KHG treatment led to decreased symptoms in 80.4% of patients with acute or chronic pharyngitis who were experiencing pharyngeal discomfort and hoarseness with no side effects.¹⁵ Further, each KHG ingredient has demonstrated multiple pharmacological activities. For instance, forsythia fruit,²² platycodon foot,²³ glycyrrhiza,²⁴ and amomum seed²⁵ inhibit pro-inflammatory cytokines such IL-1 β , interleukin-6 (IL-6), COX-2, and TNF- α , while

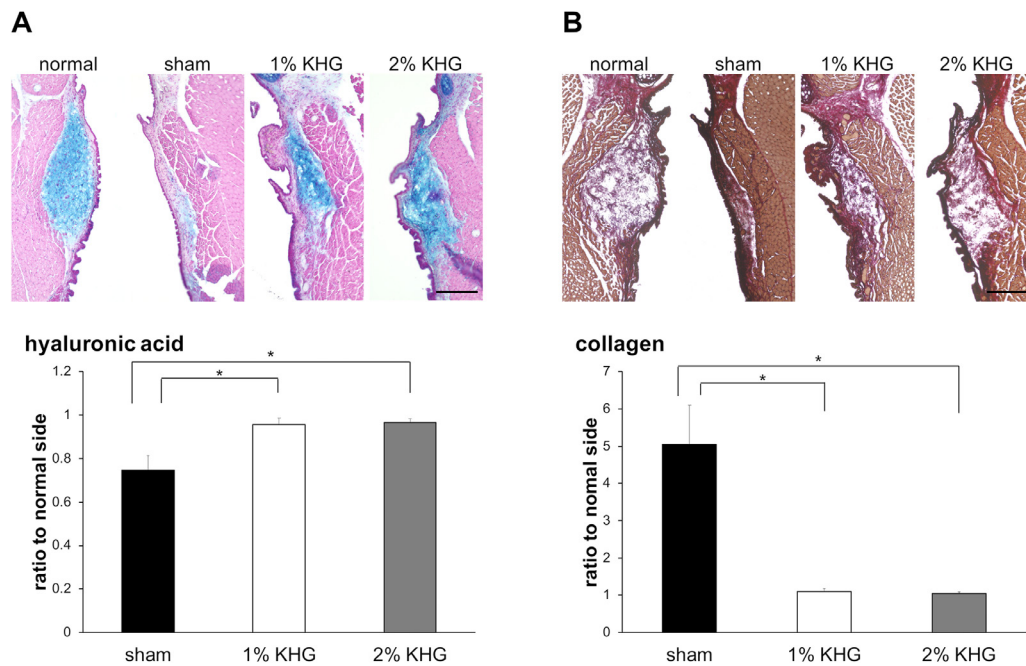


FIGURE 3. (A) Representative images of AB staining of rat vocal folds from a normal rat (uninjured side), sham group, 1% KHG group, and 2% KHG group on post injury day 56. The normalized ratio of the HA-stained area is significantly increased in the 1% and 2% KHG groups compared to the sham group. (B) Representative images of EVG staining of rat vocal fold sections from a normal rat (uninjured side), sham group, 1% KHG group, and 2% KHG group on post injury day 56. The normalized ratio of the collagen area is significantly reduced in the 1% and 2% KHG groups compared to the sham group. $n = 6$ per group, $*P < 0.05$. Data are expressed as mean $\pm$ SEM. Scale bar = 200 μ m. AB, Alcian blue; EVG, Elastica van Gieson; HA, hyaluronic acid; KHG, kyoseihatekigan.

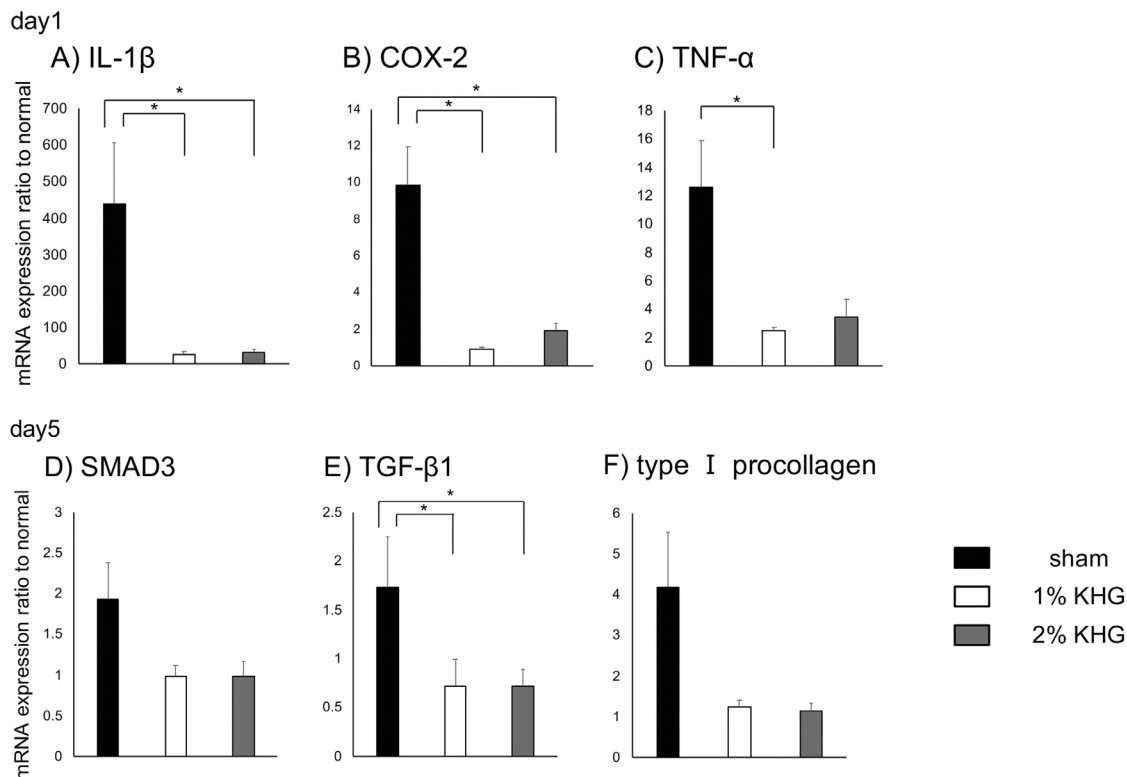


FIGURE 4. Analysis of gene expression of pro-inflammatory cytokines on post injury day 1 (A–C) and TGF- β 1 signaling on post injury day 5 (D–F) using quantitative real-time polymerase chain reaction from vocal fold tissue for each group. All genes expression levels were normalized to expression levels of SDHA and expressed as fold expression relative to a normal vocal fold. Cytokines analyzed are (A) IL-1 β , (B) COX-2, (C) TNF- α , (D) SMAD3, (E) TGF- β 1, and (F) type I procollagen. For IL-1 β , COX-2 and TGF- β 1, significant differences are observed between the sham group and the 1% and 2% KHG groups. $n = 6$ per group, $*P < 0.05$. Data are expressed as mean $\pm$ SEM. COX-2, cyclooxygenase-2; IL- β , interleukin beta; SDHA, succinate dehydrogenase complex, subunit A; TGF- β 1, transforming growth factor beta 1; TNF- α , tumor necrosis factor alpha.

gambir,²⁶ myrobalan fruit,²⁷ mentha herb,²⁸ and glycyrrhiza²⁹ exert antioxidant activity.

It has been confirmed that the acute inflammatory response and ROS play key roles in triggering fibrosis.^{7,30} The previous studies have observed the maximum cell infiltration on day 3 after injury of the vocal folds,¹⁶ and accumulation of a large amount of ROS until day 3.¹⁷ The present study also confirmed that vocal fold injury caused increases of cell infiltration and the 4-HNE immunopositive areas on post injury day 3, while KHG significantly reduced cell infiltration and 4-HNE immunopositive areas. The qPCR examination also showed that KHG significantly suppressed gene expression of pro-inflammatory cytokines including IL-1 β , COX-2, and TNF- α as compared to sham. Pro-inflammatory cytokines are molecular signals that coordinate the inflammatory response. Exaggerated or prolonged inflammation induces overexpression of these cytokines, which is linked with fibrosis.⁷ In fact, the transient overexpression of IL-1 β , accompanied by an increase of TNF- α , contributed to progressive fibrotic changes in the lung with a subsequent increase of TGF- β 1, which is an upstream driver of fibrosis.³¹ Therefore it is suggested that the early effects of KHG to suppress infiltrates of cells and proinflammatory cytokines during early phase of vocal fold wound healing are beneficial to prevent fibrosis in the following late phase.

HA is regarded as another critical element that contributes in scarless wound healing. It is well known that fetus wound healing is scarless possibly due to the abundant presence of HA in fetal tissues. An unfavorable event during vocal fold wound healing is the temporal decrease of HA in the LP after the injury.¹⁸ As it was also reported that HA injection into the vocal folds immediately after injury diminished fibrotic response,³² it should be beneficial to retain the deposition of HA during the early phase of wound healing. The current study indicated that both 1% and 2% KHG treatments significantly increased HA on post injury day 3 as compared to sham. One of the reasons why HA was well maintained may be due to suppression of ROS, because HA was reported to be susceptible to degradation by ROS.³³

KHG histologically showed prevention of fibrosis with significant HA restoration and decreased collagen deposition compared to the sham group on day 56. The qPCR study demonstrated significant suppression of gene expression of TGF- β 1, the key molecule of fibrosis, on day 5. It has been revealed that the TGF- β 1 mRNA is elevated from post injury day 3 to 14, with a prominent peak on day 7,³⁴ and SMAD3 mRNA, which is downstream of TGF- β 1, increased between post injury day 4 and 7.³⁵ Consistent with these results, our study showed increase of the mRNA expression of SMAD3 and TGF- β 1 on day 5 in the sham group, while KHG suppressed mRNA of TGF- β 1. Although the suppression of mRNA expression of SMAD3 did not reach significant level, the suppression of TGF- β 1 and its related molecules during the early phase is thought to be important for prevention of fibrosis. One of the reasons of decrease of TGF- β 1 expression may be due to the

suppression of inflammation and ROS, because previous studies reported that suppression of both inflammation and ROS decreased TGF- β 1 expression and diminished subsequent fibrotic responses.^{6,36}

The deposition of extracellular matrix during the early phase of wound healing is thought to provide a temporary scaffold for cell infiltration.^{37,38} This study showed an increase in the mRNA expression of type I procollagen at day 5 in the sham group. A previous study in rats showed that type I collagen appeared in injured vocal folds by day 3, peaked at day 5, and thereafter decreased,^{39,40} which is consistent with our results. On the other hand, KHG treatment tended to suppress type I procollagen mRNA at day 5 and resulted in better wound healing at day 56 compared to sham control. Another study reported that injection of adipose-derived mesenchymal stromal cells into the injured vocal folds of rats downregulated type I collagen mRNA at day 3 and prevented fibrosis at day 56.⁴¹ Whether or not long-term vocal fold wound healing is influenced by early-phase deposition of type I collagen remains unknown; however, this study suggests that this healing may not be affected by the downregulation of type I collagen by KHG at day 5. Since one role of early collagen deposition is to contribute to temporary scaffold formation to support cell infiltration, this deposition may not impact long-term wound healing because other molecules such as fibronectin are regarded as more important components of the early scaffold.³⁷

The current study did not address the detailed mechanism of TGF- β 1 signaling during the anti-fibrotic process associated with KHG, and therefore could not prove that KHG had a direct anti-fibrotic effect. The anti-inflammatory and antioxidant effects of KHG could indirectly prevent scarring after vocal fold injury. Further research is necessary to determine whether KHG has direct anti-fibrotic effects. It may also be needed to explore the optimal timing and duration of KHG administration; that said, In the future, it is possible that these type of agents could prove clinically beneficial during or after laryngeal surgery.

CONCLUSION

The current study showed that the application of kyoseihatekigan (KHG) during early phase of vocal fold wound healing in a rat caused decrease of cell infiltration and ROS with maintenance of HA, and decrease of gene expression of pro-inflammatory cytokines and TGF- β 1 in the early phase of wound healing. KHG prevented long-term scar formation on the vocal folds, possibly by suppressing inflammation and oxidative stress during the early phase. The control of inflammation and oxidative stress is important to prevent scar of the vocal folds, and KHG can be a novel therapeutic option for preventing vocal fold fibrosis.

Acknowledgments

The authors thank Mr. Kengo Yoshii for his assistance with statistical analysis and Ms. Keiko Izumi and Mr. Akira

Yoshimura for their technical advice regarding histological examinations.

REFERENCES

- Kumai Y. Pathophysiology of fibrosis in the vocal fold: current research, future treatment strategies, and obstacles to restoring vocal fold pliability. *Int J Mol Sci.* 2019;20(10):2551.
- Ohno S, Hirano S, Kanemaru S, et al. Transforming growth factor beta3 for the prevention of vocal fold scarring. *Laryngoscope.* 2012;122:583–589.
- Coutinho AE, Chapman KE. The anti-inflammatory and immunosuppressive effects of glucocorticoids, recent developments and mechanistic insights. *Mol Cell Endocrinol.* 2011;335:2–13.
- Samson LD. A target to suppress inflammation. *Science.* 2018;362:748–749.
- Sheppard D. Transforming growth factor beta: a central modulator of pulmonary and airway inflammation and fibrosis. *Proc Am Thorac Soc.* 2006;3:413–417.
- Cui Y, Robertson J, Maharaj S, et al. Oxidative stress contributes to the induction and persistence of TGF-beta1 induced pulmonary fibrosis. *Int J Biochem Cell Biol.* 2011;43:1122–1133.
- Wynn TA, Ramalingam TR. Mechanisms of fibrosis: therapeutic translation for fibrotic disease. *Nat Med.* 2012;18:1028–1040.
- Ingle JW, Helou LB, Li NY, et al. Role of steroids in acute phonotrauma: a basic science investigation. *Laryngoscope.* 2014;124:921–927.
- Mizuta M, Hirano S, Hiwatashi N, et al. Effect of astaxanthin on vocal fold wound healing. *Laryngoscope.* 2014;124:E1–E7.
- Li X, Zhu L, Wang B, et al. Drugs and targets in fibrosis. *Front Pharmacol.* 2017;8:855.
- Liu C, Yuan X, Tao L, et al. Xia-yu-xue decoction (XYXD) reduces carbon tetrachloride (CCl4)-induced liver fibrosis through inhibition hepatic stellate cell activation by targeting NF-kappaB and TGF-beta1 signaling pathways. *BMC Complement Altern Med.* 2015;15:201.
- Veilleux MP, Moriyama S, Yoshioka M, et al. A review of evidence for a therapeutic application of traditional Japanese Kampo medicine for oral diseases/disorders. *Med (Basel).* 2018;5(2):35.
- Yu F, Takahashi T, Moriya J, et al. Traditional Chinese medicine and Kampo: a review from the distant past for the future. *J Int Med Res.* 2006;34:231–239.
- Kimura I, Kagawa S, Nishimura K. Self-administered rhubarb-free kyoseihatekigan has novel anti-asthma effects: two case reports. *Traditional Kampo Med.* 2017;4:58–61.
- Takeno Y. Clinical effect of STR-1 on throat discomfort and hoarseness. *J New Rem Clin.* 1992;41:2311–2316.
- Branski RC, Verdolini K, Rosen CA, et al. Acute vocal fold wound healing in a rabbit model. *Ann Otol Rhinol Laryngol.* 2016;114:19–24.
- Mizuta M, Hirano S, Ohno S, et al. Expression of reactive oxygen species during wound healing of vocal folds in a rat model. *Ann Otol Rhinol Laryngol.* 2012;121:804–810.
- Tateya T, Tateya I, Sohn JH, et al. Histological study of acute vocal fold injury in a rat model. *Ann Otol Rhinol Laryngol.* 2006;115:285–292.
- Kawai S, Takagi Y, Kaneko S, et al. Effect of three types of mixed anesthetic agents alternate to ketamine in mice. *Exp Anim.* 2011;60:481–487.
- Chang Z, Ling C, Yamashita M, et al. Microarray-driven validation of reference genes for quantitative real-time polymerase chain reaction in a rat vocal fold model of mucosal injury. *Anal Biochem.* 2010;406:214–221.
- Dharmananda S. *Treatment of throat and voice disorders with Chinese medicine.* ITM; 2002.
- Jung HW, Mahesh R, Lee JG, et al. Pinorensinol from the fruits of *Forstia koreana* inhibits inflammatory responses in LPS-activated microglia. *Neurosci Lett.* 2010;480:215–220.
- Kim JY, Hwang YP, Kim DH, et al. Inhibitory effect of the saponins derived from roots of *Platycodon grandiflorum* on carrageenan-induced inflammation. *Biosci Biotechnol Biochem.* 2006;70:858–864.
- Gong G, Yuan LB, Hu L, et al. Glycyrrhizin attenuates rat ischemic spinal cord injury by suppressing inflammatory cytokines and HMGB1. *Acta Pharmacol Sin.* 2012;33:11–18.
- Choi YA, Choi JK, Jang YH, et al. Antiinflammatory effect of Amomum xanthioides in a mouse atopic dermatitis model. *Mol Med Rep.* 2017;16:8964–8972.
- Sazwi NN, Nalina T, Abdul Rahim ZH. Antioxidant and cytoprotective activities of Piper betle, Areca catechu, Uncaria gambir and betel quid with and without calcium hydroxide. *BMC Complement Altern Med.* 2013;13:351.
- Bag A, Kumar Bhattacharyya S, Kumar Pal N, et al. Anti-inflammatory, anti-lipid peroxidative, antioxidant and membrane stabilizing activities of hydroalcoholic extract of *Terminalia chebula* fruits. *Pharm Biol.* 2013;51:1515–1520.
- Trevisan SCC, Menezes APP, Barbalho SM, et al. Properties of mentha piperita: a brief review. *World J Pharm Med Res.* 2017;3:309–313.
- Asl MN, Hosseinzadeh H. Review of pharmacological effects of *Glycyrrhiza* sp. and its bioactive compounds. *Phytother Res.* 2008;22:709–724.
- Kai H, Mori T, Tokuda K, et al. Pressure overload-induced transient oxidative stress mediates perivascular inflammation and cardiac fibrosis through angiotensin II. *Hypertens Res.* 2006;29:711–718.
- Kolb M, Margetts PJ, Anthony DC, et al. Transient expression of IL-1beta induces acute lung injury and chronic repair leading to pulmonary fibrosis. *J Clin Invest.* 2001;107:1529–1536.
- Hansen JK, Thibault SL, Walsh JF, et al. In vivo engineering of the vocal fold extracellular matrix with injectable hyaluronic acid hydrogels: early effects on tissue repair and biomechanics in a rabbit model. *Ann Otol Rhinol Laryngol.* 2005;114:662–670.
- Sato H, Takahashi T, Ide H, et al. Antioxidant activity of synovial fluid, hyaluronic acid, and two subcomponents of hyaluronic acid. Synovial fluid scavenging effect is enhanced in rheumatoid arthritis patients. *Arthritis Rheumatism.* 1988;31:63–71.
- Ohno T, Hirano S, Rousseau B. Gene expression of transforming growth factor-beta1 and hepatocyte growth factor during wound healing of injured rat vocal fold. *Laryngoscope.* 2009;119:806–810.
- Paul BC, Rafii BY, Gandonu S, et al. Smad3: an emerging target for vocal fold fibrosis. *Laryngoscope.* 2014;124:2327–2331.
- Yao QY, Xu BL, Wang JY, et al. Inhibition by curcumin of multiple sites of the transforming growth factor-beta1 signalling pathway ameliorates the progression of liver fibrosis induced by carbon tetrachloride in rats. *BMC Complement Altern Med.* 2012;12:156.
- Hirano S, Bless DM, Rousseau B, et al. Fibronectin and adhesion molecules on canine scarred vocal folds. *Laryngoscope.* 2003;113:966–972.
- Hiwatashi N, Hirano S, Mizuta M, et al. The efficacy of a novel collagen-gelatin scaffold with basic fibroblast growth factor for the treatment of vocal fold scar. *J Tissue Eng Regen Med.* 2017;11:1598–1609.
- Tateya T, Tateya I, Sohn JH, et al. Histologic characterization of rat vocal fold scarring. *Ann Otol Rhinol Laryngol.* 2005;114:183–191.
- Lim X, Tateya I, Tateya T, et al. Immediate inflammatory response and scar formation in wounded vocal folds. *Ann Otol Rhinol Laryngol.* 2006;115:921–929.
- Morisaki T, Kishimoto Y, Tateya I, et al. Adipose-derived mesenchymal stromal cells prevented rat vocal fold scarring. *Laryngoscope.* 2018;128:E33–E40.